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EDITORIAL

Ciraparantag as a potential universal anticoagulant reversal agent

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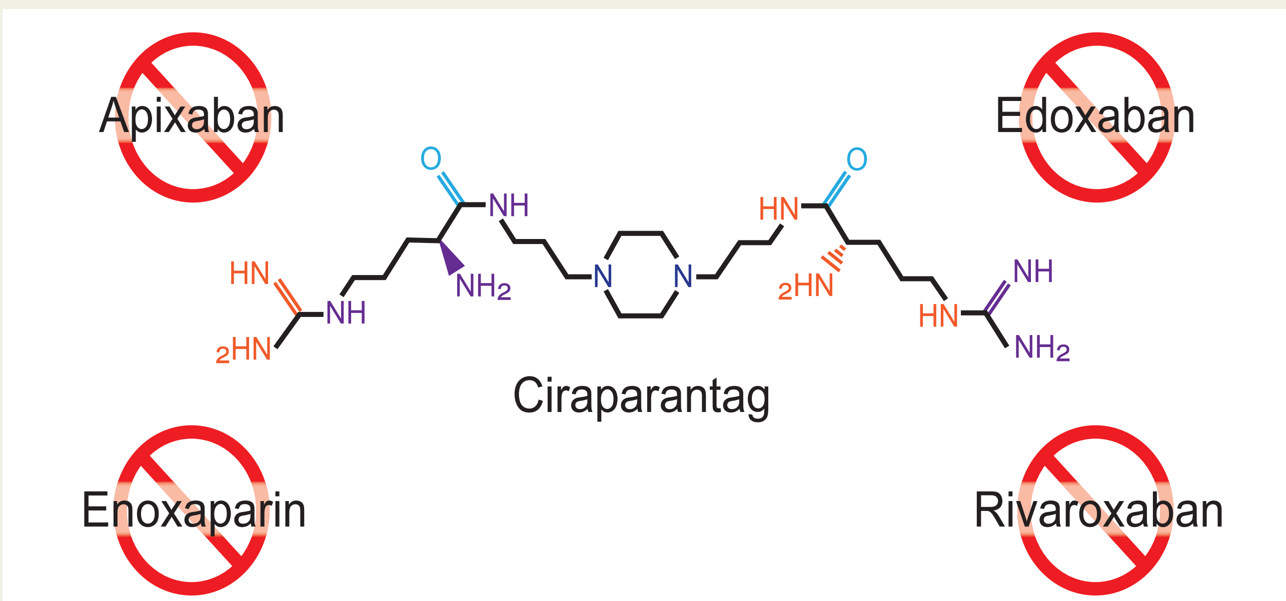
This editorial refers to 'Ciraparantag reverses the anticoagulant activity of apixaban and rivaroxaban in healthy elderly subjects' by J. Ansell *et al.*, doi:10.1093/eurheartj/ehab637.



Graphical Abstract Ciraparantag binds to apixaban, edoxaban, enoxaparin, and rivaroxaban and reverses their anticoagulant activity.

Direct oral anticoagulants (DOACs) include dabigatran, which inhibits thrombin, and apixaban, edoxaban, and rivaroxaban, which inhibit factor Xa (FXa). Use of the DOACs increased when guidelines gave preference to them over vitamin K antagonists (VKAs) such as warfarin for stroke prevention in patients with non-valvular atrial fibrillation and for treatment of venous thrombo-embolism.[1–3] Of the DOACs, apixaban and rivaroxaban are the most widely used.[4] Although the DOACs are associated with less serious bleeding than VKAs, particularly less intracranial haemorrhage, bleeding remains the major side effect of the DOACs. Consequently, agents that rapidly reverse the anticoagulant effects of the DOACs could streamline management and improve outcomes of patients presenting with serious bleeding or requiring urgent interventions or surgery.

Idarucizumab, an antibody fragment that binds dabigatran with high affinity, is widely available for rapid dabigatran reversal.[5] Andexanet, a recombinant FXa variant, is licensed in Europe and the USA for reversal of apixaban and rivaroxaban.[6] However, andexanet is not approved in Canada and the UK, is only available in ∼12% of hospitals

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Table 1 Reversal agents for anticoagulants

	Ciraparantag	Andexanet[a]	Four-factor PCC[b]	Idarucizumab[b]
Structure	Synthetic water-soluble cationic small molecule	Inactive, Gla-domain-truncated, recombinant factor Xa	Contains prothrombin and factors VII, IX, and X	Humanized monoclonal antibody fragment
Molecular weight (Da)	512	39 000	50 000–72 000	47 766
Anticoagulants reversed	Direct oral anticoagulants (apixaban, dabigatran, edoxaban, and rivaroxaban) and heparin	Direct oral factor Xa inhibitors (apixaban, edoxaban and rivaroxaban) and heparin	Direct oral anticoagulants (apixaban, dabigatran, edoxaban, and rivaroxaban) and vitamin K antagonists	Dabigatran
Mechanism of action	Binds to direct oral anticoagulants and heparin via non-covalent hydrogen bonds and charge–charge interactions	Competitive binding to oral factor Xa inhibitors and competition with factor Xa for binding to heparin-catalysed antithrombin	Enhances factor Xa and thrombin generation	Binds free and thrombin-bound dabigatran
Administration	Single i.v. bolus	I.v. bolus followed by a 2 h infusion	I.v. infusion over 10–20 min	Single or double i.v. bolus
Storage	Room temperature	Refrigerated	Refrigerated or room temperature	Refrigerated
Cost	Probably low	Very high	Moderate	Moderate

Gla, gamma carboxyglutamic acid; PCC, prothrombin complex concentrate.
[a]Licensed in the USA and Europe,
[b]Off-label use.
[c]Licensed in the USA, Europe, and many other jurisdictions.

in the USA,[7] and enthusiasm for its use is limited by its high cost. Instead, four-factor prothrombin complex concentrate (PCC) is often used for urgent reversal of apixaban or rivaroxaban based on data from cohort studies suggesting that it is effective.[8] Therefore, there remains a need for effective and relatively low-cost reversal agents for oral FXa inhibitors.

Ciraparantag, a small water-soluble synthetic molecule, has the potential to fill this gap (see *Table 1*).[9] By binding to DOACs and heparin via non-covalent hydrogen bonds and charge–charge interactions, ciraparantag reverses their anticoagulant effects by displacing them from FXa and thrombin.[9] Previous studies in healthy volunteers have shown that ciraparantag reverses the anticoagulant effects of edoxaban and enoxaparin.[10,11]

In this issue of the *European Heart Journal*, Ansell and colleagues report the results of two dose-ranging trials of ciraparantag for reversal of apixaban or rivaroxaban in healthy subjects between the ages of 50 and 75 years.[12] Participants in the first study received apixaban (10 mg twice daily) for 3.5 days, while those in the second study received rivaroxaban (20 mg once daily) for 3 days. The anticoagulant effect of apixaban and rivaroxaban at steady state was assessed with the whole blood clot time (WBCT). Subjects whose WBCT was prolonged ≥20% over baseline with apixaban and ≥25% with rivaroxaban were randomized in a 3:1 ratio to ciraparantag or placebo given as a single intravenous bolus. Ciraparantag doses of 30, 60, or 120 mg were evaluated for apixaban reversal, while ciraparantag doses of 30, 60, 120, or 180 mg were evaluated for rivaroxaban reversal. In both studies, the primary efficacy outcome was complete reversal, which was defined as a reduction in the WBCT to ≤10% of baseline at 15, 30, 45, and 60 min and at 3, 5, and 24 h after ciraparantag or placebo administration.

With ciraparantag doses of 30, 60, or 120 mg, there was reversal of apixaban in 67, 100, and 100% of participants, respectively, compared with reversal in 17% of those given placebo. Likewise, with ciraparantag doses of 30, 60, 120, and 180 mg, there was reversal of rivaroxaban in 58, 75, 67, and 100% of participants, respectively, compared with reversal in 13% of those given placebo. Reversal was observed within 15 min of ciraparantag administration and was sustained for up to 24 h. Although adverse events with ciraparantag were reported in ∼40% of participants, these were mostly mild and limited to transient hot flushes or a feeling of warmth or coldness with the infusion.

The strengths of these studies include the dosing of rivaroxaban and apixaban to steady state prior to reversal, the inclusion of older volunteers, the randomized placebo-controlled design, and the comprehensive WBCT sampling over 24 h. However, like many phase II studies, the interpretation of the data is limited by the small sample size, the inclusion of healthy volunteers with normal renal function who are unlikely to be representative of patients treated in clinical practice, and the use of the WBCT as a surrogate for efficacy. The WBCT was used in place of conventional coagulation assays because ciraparantag binds to the sodium citrate used for blood collection and to the reagents used to trigger clotting in these tests. Unfortunately, the WBCT is not routinely available for clinical use and appears to be relatively insensitive to the anticoagulant effects of apixaban and, to a lesser extent, rivaroxaban. Thus, at steady state, the WBCT was not prolonged over baseline in 9 out of 60 volunteers

(15%) given apixaban and in 2 out of 69 volunteers (3%) given rivaroxaban. The findings with apixaban are particularly noteworthy because subjects were given 10 mg twice daily, which is double the usual dose of apixaban administered for chronic anticoagulation. Despite these drawbacks, however, the current studies add to the evidence that ciraparantag has potential as a universal reversal agent by showing that it rapidly and completely reverses the anticoagulant effects of apixaban and rivaroxaban.

The next and the critical step in ciraparantag development is to show that apixaban or rivaroxaban reversal with ciraparantag improves clinical outcomes in patients who present with serious bleeding or require urgent intervention or surgery. Such a study will face several challenges. First, the limited availability of the WBCT will render it difficult to monitor the response to ciraparantag. Second, with the current use of four-factor PCC or andexanet for apixaban and rivaroxaban reversal, a cohort study design as was used for approval of idarucizumab and andexanet may no longer be accepted by regulatory agencies. Instead, a trial comparing ciraparantag with standard of care may be mandated. If four-factor PCC or andexanet is used as standard of care, it will be difficult to show superiority of ciraparantag because both four-factor PCC and andexanet have been reported to restore haemostasis in ~80% of cases.[8] Although demonstration of non-inferiority may be sufficient to gain approval, use of ciraparantag will then depend on its availability, cost, and side effect profile. Third, if patients requiring urgent intervention or surgery are included in such a trial, those needing heparin will have to be excluded because like andexanet, ciraparantag also reverses heparin.

Although the road forward is challenging and much work remains to be done, the study by Ansell and colleagues provides the impetus to study ciraparantag reversal in patients taking apixaban or rivaroxaban.

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